

Deep learning-based coronary artery motion estimation and compensation for short-scan cardiac CT

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Purpose: During a typical cardiac short scan, the heart can move several millimeters. As a result, the corresponding CT reconstructions may be corrupted by motion artifacts. Especially the assessment of small structures, such as the coronary arteries, is potentially impaired by the presence of these artifacts. In order to estimate and compensate for coronary artery motion, this manuscript proposes the deep partial angle-based motion compensation (Deep PAMoCo).

Methods: The basic principle of the Deep PAMoCo relies on the concept of partial angle reconstructions (PARs), that is, it divides the short scan data into several consecutive angular segments and reconstructs them separately. Subsequently, the PARs are deformed according to a motion vector field (MVF) such that they represent the same motion state and summed up to obtain the final motion-compensated reconstruction. However, in contrast to prior work that is based on the same principle, the Deep PAMoCo estimates and applies the MVF via a deep neural network to increase the computational performance as well as the quality of the motion compensated reconstructions.

Results: Using simulated data, it could be demonstrated that the Deep PAMoCo is able to remove almost all motion artifacts independent of the contrast, the radius and the motion amplitude of the coronary artery. In any case, the average error of the CT values along the coronary artery is about 25 HU while errors of up to 300 HU can be observed if no correction is applied. Similar results were obtained for clinical cardiac CT scans where the Deep PAMoCo clearly outperforms state-of-the-art coronary artery motion compensation approaches in terms of processing time as well as accuracy.

Conclusions: The Deep PAMoCo provides an efficient approach to increase the diagnostic value of cardiac CT scans even if they are highly corrupted by motion. © 2021 The Authors. *Medical Physics* published by Wiley Periodicals LLC on behalf of American Association of Physicists in Medicine [https://doi.org/10.1002/mp.14927]

Key words: cardiac CT, deep learning, motion artifact correction, motion compensation, motion estimation

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1. INTRODUCTION

Cardiac imaging is playing an increasingly important role in the detection and diagnosis of cardiac diseases.¹ In this

context, cardiac CT has become a promising imaging modality to provide high-resolution reconstructions of the heart in a noninvasive manner.² Thus, several applications, most of which are related to postsurgical follow-up or diagnosis of

coronary artery diseases via coronary CT angiography, have found their way into clinical practice.^{3,4}

In order to account for cardiac motion, these application usually rely on ECG-gated scan protocols that synchronize the CT scan with an ECG signal. In that way, the reconstruction can be restricted to a short scan ($180^\circ + \text{fan angle}$) corresponding to a certain interval of the cardiac cycle, preferably the one with the least motion (e.g., the mid-diastole or the end-systole⁵). For the sake of minimizing radiation dose, prospective gating strategies also restrict the data acquisition to that interval by triggering the x-ray exposure according to the ECG.⁶ However, compared to retrospective approaches that acquire data continuously, low and stable heart rates are required to ensure an appropriate timing of the acquisition window. To some extent this prerequisite can be achieved by pre-procedure medications to lower the heart rate. Therefore, beta-blockers are frequently administered prior to cardiac CT examinations.⁵

Although ECG-gated scan protocols, best phase selection as well as heart rate control drugs allow to reduce the impact of cardiac motion on image quality, the temporal resolution of typical single source CT systems is sometimes still too low to avoid motion artifacts completely. In particular, this holds true for prospectively gated scans with an inappropriate timing of the acquisition window or for patients with high heart rates and contraindications to beta-blockade. As a result, the assessment of small structures such as coronary arteries may be impaired in these cases. Therefore, several efforts have been made to further increase the temporal resolution or to correct motion artifacts retrospectively. Besides dual-source CT,⁷ which is probably the most effective approach to increase the temporal resolution on the hardware side, several software based approaches have been proposed.

One option is to limit the angular coverage of the CT acquisition to less than what is required for an exact analytical reconstruction, that is, to less than the short scan range ($180^\circ + \text{fan angle}$). Since this allows to reduce the scan time, the temporal resolution is increased, however, at cost of the reconstruction problem becoming ill-posed. Thus, it needs to be solved iteratively incorporating some sort of prior knowledge to suppress limited angle artifacts.^{8–11} As the scan range can only be reduced to a certain extent, the temporal resolution cannot be increased by more than about 30%.¹⁰

More sophisticated approaches are based on motion compensation (MoCo). In general, these approaches aim to determine a motion vector field (MVF) that models cardiac motion and can be used to compensate for it during or subsequent to the reconstruction. Here, one prominent strategy is to acquire and to reconstruct multiple phases of the cardiac cycle. Using an appropriate registration algorithm, the MVF is determined as the transformation that maps these phases to a certain reference phase.^{12–15} However, a disadvantage of these registration-based approaches is that they highly depend on the quality of the reference phase. Furthermore, they may not use the x-ray dose optimally, especially if only one motion phase is of interest.

In this case, a more dose-efficient approach is to use data only from a single short scan acquisition of that particular phase. To do so, Rohkohl *et al.* proposed to incorporate a motion model in the corresponding short scan reconstruction and to optimize its parameters such that a motion artifact metric (MAM) in image domain is minimized.¹⁶ It has been shown that this approach can efficiently improve the image quality of the best phase, that is, the cardiac phase with the least motion, but may fail for cardiac phases with higher motion amplitudes due to local minima of the MAM.

To enable a more robust motion compensation, which applies to any cardiac phase, the concept of partial angle reconstructions (PAR) has been introduced.¹⁷ Its basic principle is to divide the short scan into small angular segments and to reconstruct them separately. Although these so-called PARs suffer from severe limited angle artifacts, they have a high temporal resolution and thus allow for an accurate estimation of motion. Kim *et al.* proposed to do so by estimating a MVF via nonrigid registration from two opposing PARs that are separated by 180° .¹⁸ Subsequently, the MVF for any view in between is approximated by linear interpolation and included in the reconstruction to derive motion-compensated CT images. While this approach is particularly effective for estimating motion perpendicular to the projection direction of the two PARs, the estimate of the parallel motion component is less reliable. Therefore, Hahn *et al.* proposed the so-called partial angle-based motion compensation (PAMoCo) that uses all the PARs to estimate and compensate for motion.¹⁹ Here, the PAMoCo makes use of the MAM concept, that is, it determines a MVF that is applied to the PARs such that their sum minimizes the image entropy. However, to avoid local minima of the entropy, the PAMoCo is based on a complex optimization strategy which requires about 15 min to process a single coronary artery.¹⁹

More recent work started to exploit the potential of deep learning. Here, existing approaches either train neural networks to predict coronary artery motion²⁰ or to correct the corresponding motion artifacts post to the reconstruction.²¹ While the former class of approaches requires to include the trained network into a certain motion compensation pipeline, the latter rather performs an image-based postprocessing that is not constrained to model actual cardiac motion.

In order to enable an end-to-end training of coronary artery motion compensation, this study proposes the Deep PAMoCo. It follows the basic strategy of the PAMoCo, but replaces the determination as well as the application of the MVF by a deep neural network. In particular, this is achieved by combining an encoder network, which is trained to estimate the parameters of a motion model, with a spatial transformer module.²² The latter provides a general framework for implementing trainable spatial transformations in neural networks and has been used successfully for motion compensation tasks in other areas of research.^{23–26} Here it is designed to compute the MVF of each input PAR according to the motion model and to apply it internally to derive the motion-compensated reconstruction. The feasibility of the Deep PAMoCo has been demonstrated by the authors in a

preliminary study for a two-dimensional (2D) application and a linear motion model.²⁷ Here, we extend the Deep PAMoCo to estimate 3D MVFs according to a more complex motion model. Furthermore, we provide a more solid evaluation for simulated and measured data that, in contrast to our previous study, demonstrates superior performance compared to the conventional PAMoCo.

2. MATERIAL AND METHODS

2.A. Deep PAMoCo

2.A.1. Basic principle

The principle of the Deep PAMoCo is depicted in Fig. 1. In a first step, an initial reconstruction f is performed. Since we aim to limit the motion compensation to the region of the coronary arteries, their position is determined based on the initial reconstruction using a segmentation software provided by Siemens. Subsequently, $2K + 1$ PARs f_k , with $-K \leq k \leq K$, are generated by forward- and backprojecting data within patches centered at the coronary artery using a parallel beam geometry. Here, each PAR corresponds to a nonoverlapping segment covering the angular interval $[\alpha_k - \Delta\alpha/2, \alpha_k + \Delta\alpha/2]$, with $\alpha_k = \alpha_0 + k \cdot \Delta\alpha$, $\Delta\alpha = \pi/(2K + 1)$, and α_0 being the central angle of the central PAR. Thus, the temporal resolution of the PARs is approximately $t_{\text{res}} = 0.5 \cdot t_{\text{rot}}/(2K + 1)$, where t_{rot} is the gantry rotation time. With a proper choice of K , the temporal resolution of the PARs is high enough to represent a quasi-static sub-phase of the cardiac cycle. Applying a MVF to the PARs that maps them to the same motion state yields the motion-compensated reconstruction as:

$$f_{\text{MoCo}}(\mathbf{r}) = \sum_{k=-K}^K f_k(\mathbf{r} + \mathbf{s}(\mathbf{r}, t_k)), \quad (1)$$

with $\mathbf{r} = (x, y, z)^T$ denoting the spatial coordinate and $\mathbf{s}(\mathbf{r}, t_k)$ denoting the MVF applied to the PAR f_k at time point t_k . In contrast to the conventional PAMoCo that estimates $\mathbf{s}(\mathbf{r}, t)$ such that the image entropy of f_{MoCo} is minimized, the Deep PAMoCo uses a neural network to determine $\mathbf{s}(\mathbf{r}, t)$ as well as to apply it according to Eq. (1). Finally, the patch is

reinserted into the initial reconstruction to obtain the final motion-compensated reconstruction.

It has to be noted that the current workflow involves a redundant forward- and filtered backprojection of the patch data in parallel beam geometry. Here, the use of the parallel beam geometry can be justified by the fact that the patches are rather small (see Fig. 1) such that the incident rays are almost parallel. Nevertheless, the PARs could also be reconstructed directly from the acquired raw data. As long as the reconstruction does not use redundant data, that is, every voxel is backprojected from only 180° , forward- and filtered backprojection implement inverse operations and both strategies are equivalent. In particular, this also holds true for scans including motion. Therefore, motion artifacts that are present in the initial reconstruction vanish during the forward projection and do not affect the quality of the PARs. The advantage of the present strategy, however, is that neither access to the raw data nor to the geometry of the CT system is required. Thus, the proposed approach can be implemented more easily, especially if data of different CT systems are to be processed. On the downside, the additional forward- and backprojection may introduce a slight blurring to the final motion-compensated images.

2.A.2. Motion model

To reduce the number of parameters to be estimated, the MVF is constrained to a custom motion model. Assuming that the motion trajectory is a smooth function in time, we model it as a polyharmonic spline function with three control points $\mathbf{s}(t_{-K})$, $\mathbf{s}(t_0)$, $\mathbf{s}(t_K)$ corresponding to the motion vectors of the first, the central, and the last PAR. Choosing cubic radial basis functions, $\mathbf{s}(\mathbf{r}, t)$ is given as:

$$\mathbf{s}(\mathbf{r}, t) = w(\mathbf{r}) \cdot \sum_i \alpha_i |t - t_i|^3 + \beta \cdot t + \gamma, \quad (2)$$

with $i \in \{-K, 0, K\}$ and $w(\mathbf{r})$ being a weighting function that forces a smooth decay of the MVF towards the patch boundaries and is exactly zero for the boundary voxels. Here, the coefficients α_i , β , and γ are determined such that the interpolant exactly fits the value of the control points at t_{-K} , t_0 , and t_K . Furthermore, we assume that the central partial angle

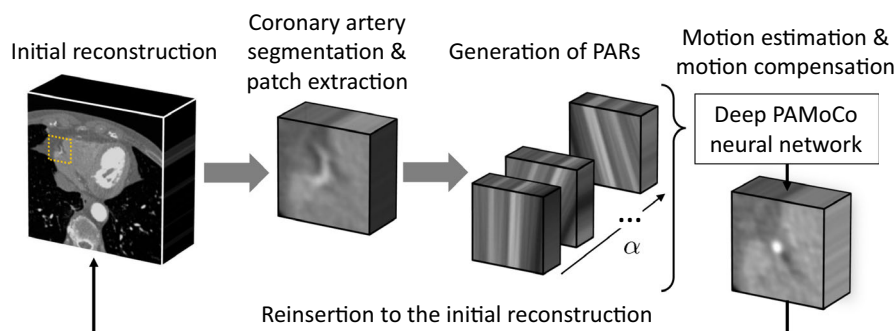


FIG. 1. Basic workflow of the Deep PAMoCo.

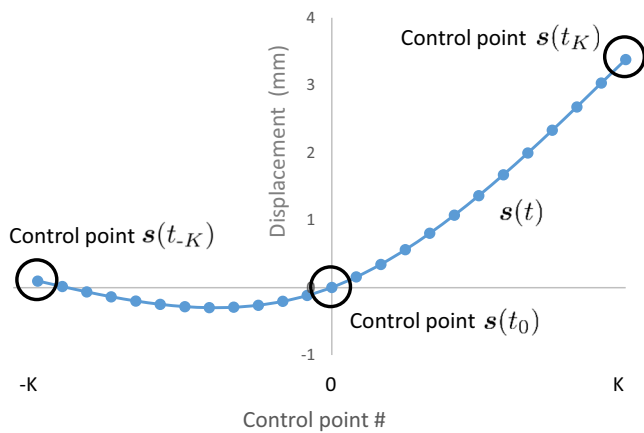


FIG. 2. Illustration of the motion model. The temporal dependency is modeled by a spline function. Here, the central control point is always set to zero. The first and the last control point, corresponding to the first and the last partial angle image, are estimated by the neural network. Any other displacement vector is determined by a polyharmonic spline interpolation. [Color figure can be viewed at wileyonlinelibrary.com]

represents the reference motion state, i.e. its control point equals the zero vector. Thus, the motion trajectory is modeled by six parameters representing the 3D coordinates of the first and the last control point. As depicted in Fig. 2, the MVF of any other partial angle image f_k is determined by evaluating Eq. (2) at the corresponding time point t_k . It has to be noted that the complete interpolation procedure is differentiable with respect to all inputs, and thus, can be included directly into a the training process of a neural network.

2.A.3. Neural network

The Deep PAMoCo network is designed to estimate the parameters of the motion model described in Section 2.A.2. To enable an end-to-end training, the motion model is applied internally to obtain the motion-compensated output. The corresponding network architecture is shown in Fig. 3. Here, the PARs of a $128 \times 128 \times 15$ patch containing the coronary artery are given as a 3D $(2K+1)$ -channel input. These PARs are then processed by an encoder-like network that repeatedly applies convolution ($3 \times 3 \times 3$ kernel, batch normalization, and ReLU activation) and pooling ($2 \times 2 \times 1$ max pooling) operations to extract features related to coronary artery motion. A set of fully connected neurons is used subsequently to map these features to the six parameters of the motion model.

Given the parameters of the motion model, a MVF $\mathbf{s}(\mathbf{r}, t)$ is computed for every input PAR according to Eq. (2) and applied according to Eq. (1) to derive the final motion-compensated output. In particular, this operation is implemented by the spatial transformer module as a resampling operation in which each output voxel is computed by applying a linear sampling kernel centered at a particular position of the input PAR. Therefore, the output is defined to be located on a regular $128 \times 128 \times 15$ grid where each voxel is addressed by the index (l, m, n) and corresponds to the vertex at $\mathbf{r}_{lmn} = (x_l, y_m, z_n)^T = (x_0 + l \cdot dx, y_0 + m \cdot dy, z_0 + n \cdot dz)^T$, with dx, dy , and dz representing the voxel spacing and x_0, y_0 , and z_0 the grid origin. The corresponding source coordinates $\mathbf{r}'_{lmn}$ in the input PAR f' define the sampling

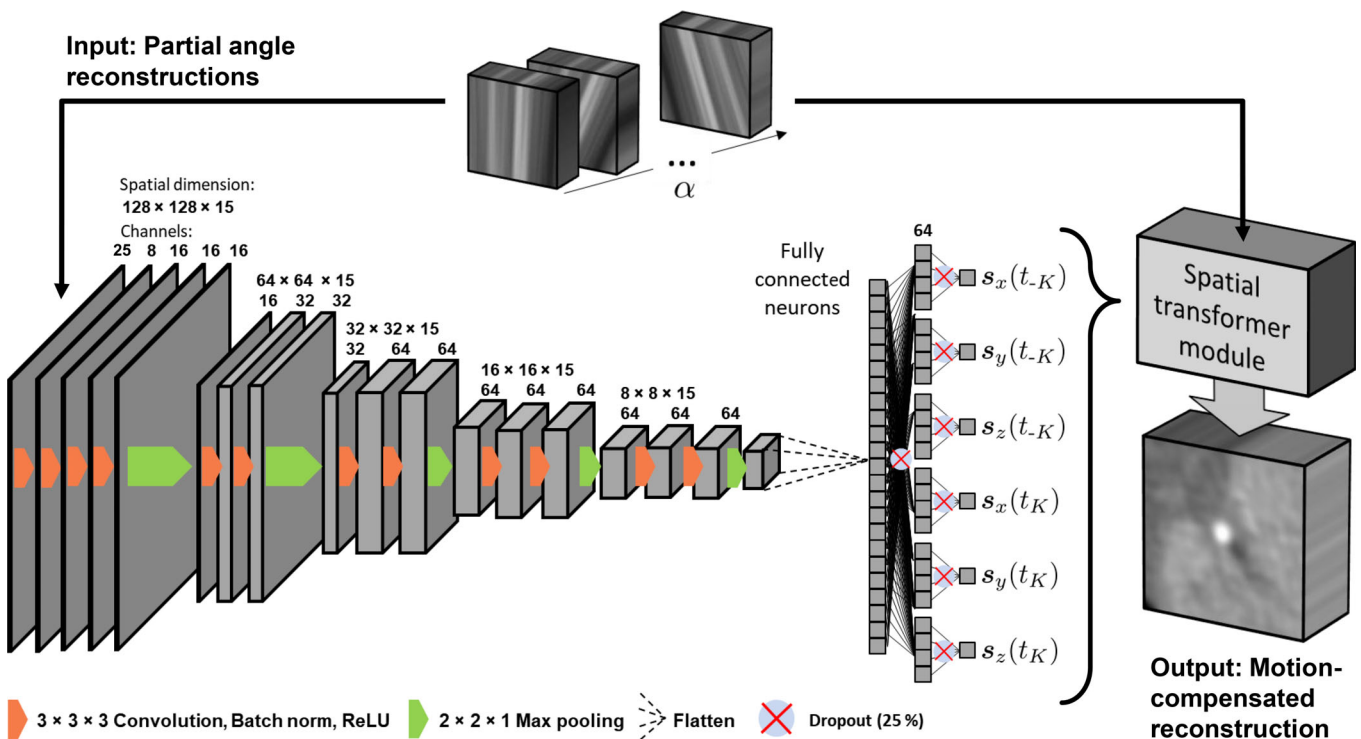


FIG. 3. Architecture of the neural network that is used to estimate the six parameters of the motion model. To enable an end-to-end training, a spatial transformer module is used that internally applies the transformation, according to the motion model, to the input partial angle images. [Color figure can be viewed at wileyonlinelibrary.com]

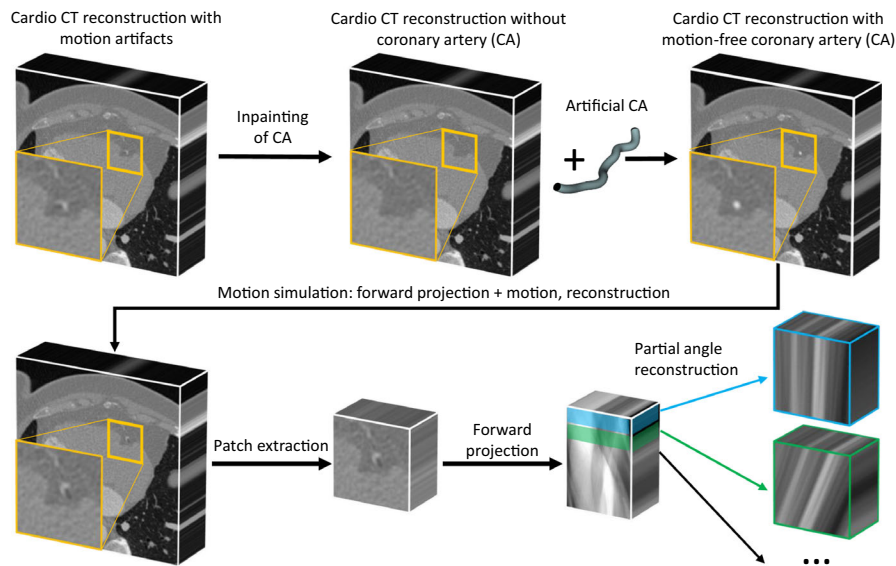


FIG. 4. Basic workflow of the training data generation. To have prior CT volumes without motion artifacts, the coronary artery (which may show motion artifacts) is removed from real CT reconstructions and replaced by an artificial one without artifacts. Subsequently, this motion artifact-free CT volume is used to simulate motion again and to generate the partial angle images that are used as input to the Deep PAMoCo network. [Color figure can be viewed at wileyonlinelibrary.com]

points and are related to $\mathbf{r}_{lmn}$ through the MVF: $\mathbf{r}'_{lmn} = \mathbf{r}_{lmn} + \mathbf{s}(\mathbf{r}_{lmn}, t_k)$. Thus, the value f_{lmn} of the transformed PAR at position $\mathbf{r}_{lmn}$ can be calculated as:

$$f_{lmn} = \sum_{l', m', n'} f'_{l'm'n'} \cdot \max\left(0, 1 - \left|\frac{x_l - x_{l'}}{dx}\right|\right) \cdot \max\left(0, 1 - \left|\frac{y_m - y_{m'}}{dy}\right|\right) \cdot \max\left(0, 1 - \left|\frac{z_n - z_{n'}}{dz}\right|\right) \quad (3)$$

2.A.4. Generation of training data

In this study, we train the Deep PAMoCo network described in Section 2.A.3 in a supervised setup. Therefore, paired training data with and without motion artifacts are required. Since such data are hard to obtain from real measurements, we generate them using CT simulations instead. As depicted in Fig. 4, our simulation consists of the following steps.

First, cardiac CT reconstructions are used to generate ideal prior volumes without coronary artery motion artifacts. As not all reconstructions that are available to us display the coronary artery sharply, it is replaced by an artificial one. To do so, the coronary artery is segmented and removed by an exemplar-based inpainting approach. Subsequently, the centerline of the segmentation is used to generate an artificial, motion artifact-free, coronary artery which is then inserted in the same place. This strategy not only allows to use any cardiac CT reconstruction as prior but also to generate several different coronary artery representations differing in shape, size, and contrast.

In the next step, CT scans are generated by a forward projection of the prior volume. To simulate coronary artery motion, the prior volume is transformed during the forward

projection according to the motion model described in Section 2.A.2. In particular, each simulation is performed by sampling random values of the two control points to cover different motion patterns. Here, the sampling is constrained such that the motion velocity of the coronary artery does not exceed 70 mm/s in the axial plane and 25 mm/s in z-direction. Since the appearance of motion artifacts also depends on the source trajectory, the start angle of the simulated CT scan is sampled randomly as well. Finally, a filtered backprojection of the simulated projection data is performed to obtain the corresponding CT reconstructions containing motion artifacts.

In a final step patches centered around the coronary artery are extracted and PARs are generated by a 180° parallel beam forward projection and a subsequent backprojection of 25 non-overlapping angular segments covering 7.2°. In any case, the PARs are reconstructed on a $128 \times 128 \times 15$ voxel grid with a voxel spacing of $0.35 \text{ mm} \times 0.35 \text{ mm} \times 0.6 \text{ mm}$.

2.A.5. Optimization

To train the Deep PAMoCo network, 100 000 samples based on 25 cardiac CT reconstructions were generated. Here, each sample corresponds to a pair of 25 PARs and their respective label, simulated with an independent set of parameters. In particular, each sample was simulated using a random start angle and a random motion pattern as described in Section 2.A.4. Furthermore, each sample was simulated with a random coronary artery radius between 0.4 and 2.1 mm as well as random coronary artery contrast between 150 and 1400 HU which corresponds to the contrast range observed for the clinical data available to us. The samples were divided into 88% training data (data of 22 prior volumes) and 12% validation data (data of the

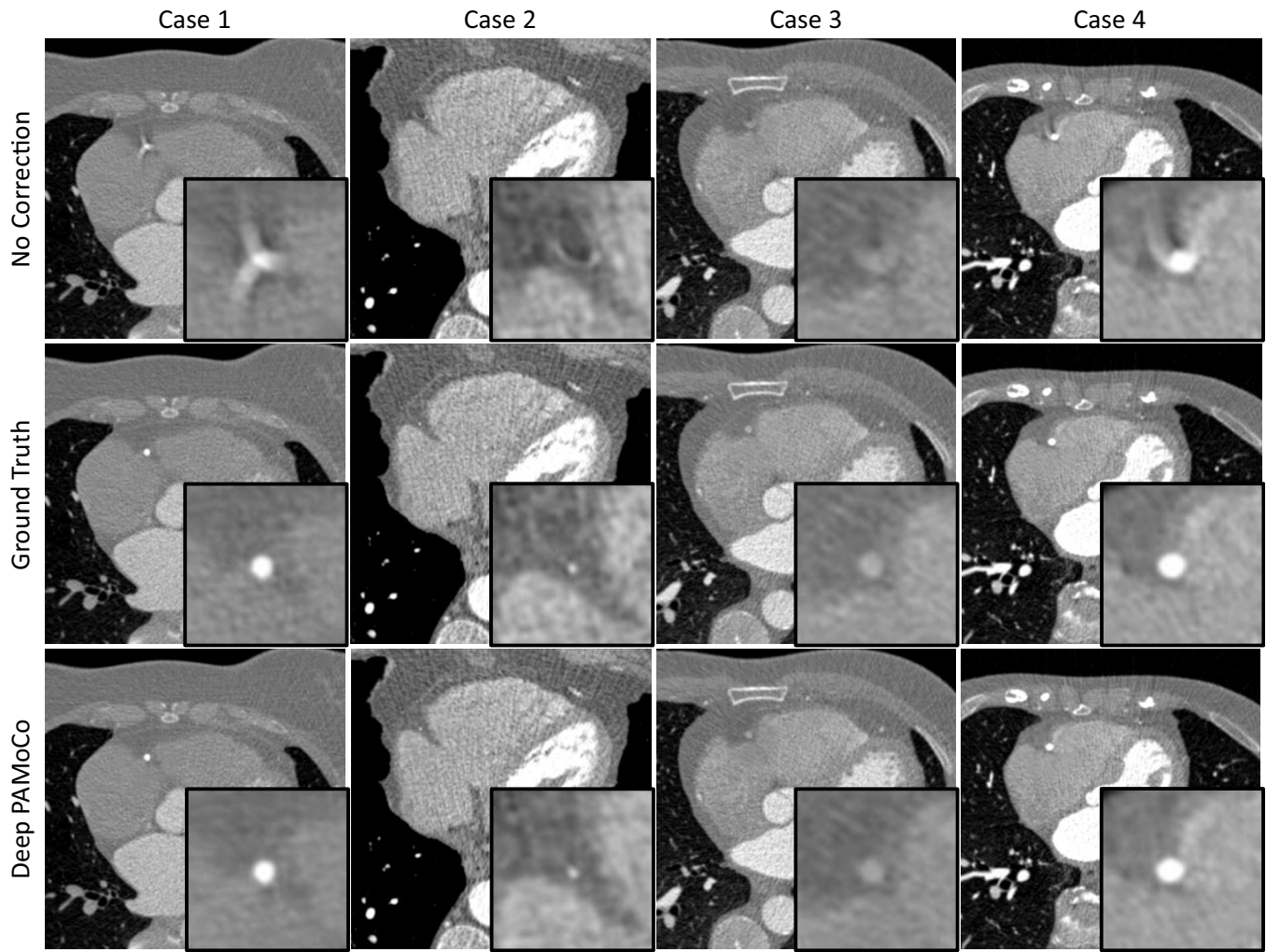


FIG. 5. Exemplary results of the simulation study without correction (top), an ideal correction (middle) as well as with the deep PAMoCo correction (bottom). $C = 0$ HU, $W = 1500$ HU.

remaining 3 prior volumes). Using these training data, the parameters of the network were optimized in a supervised setting. PARs were used as input to the network, while labels were generated by an ideal motion compensation that is based on the exact motion trajectory that was used for simulation. Here, the ideal motion compensation was preferred over the ideal prior volume to account for the slight blurring that may be caused by the resampling operation described in Section 2.A.3. The training was performed on an NVIDIA GeForce RTX 2080 Ti for 50 epochs (≈ 4.5 h/epoch) using an Adam optimizer (learning rate = 0.00025, $\beta_1 = 0.9$, $\beta_2 = 0.999$, no weight decay), a batch size of 4 and the mean squared error between the prediction and the label as loss function. After each epoch the validation loss was evaluated. Finally, the network with the minimal validation loss was used for testing (see Section 2.B).

2.B. Evaluation

The performance of the Deep PAMoCo was evaluated for simulated and measured data. Here, the simulation study is

based on ten cardiac CT reconstructions that differ from the ones that were used to generate the training and validation data. Aside from that, the simulation is similar to what is described in Section 2.A.4, that is, an artificial coronary artery is inserted, motion is simulated by deforming the prior volume during forward projection, and PARs are generated by forward and backprojection in parallel beam geometry. In total, 5000 samples were generated and evaluated by comparing the output of the Deep PAMoCo $f_{\text{DeepPAMoCo}}$ to the corresponding ground truth motion compensation f_{GT} , that is, a motion compensation based on the known motion trajectory. Here, the mean absolute error along the centerline of the coronary artery (MAE_{CA}), was used as a performance measure with.

$$\text{MAE}_{\text{CA}} = \frac{\int_C |f_{\text{GT}}(\mathbf{r}) - f_{\text{DeepPAMoCo}}(\mathbf{r})| ds}{\int_C ds}, \quad (4)$$

and C being the centerline curve.

To test the performance of the Deep PAMoCo for real measurements, 13 patient scans, conducted at a Siemens SOMATOM AS+, were evaluated. All scans were performed

with a tube voltage of 80 kV (4 patients), 100 kV (5 patients), or 120 kV (4 patients) using a standard ECG-triggered cardiac CT protocol that used an active tube current between 40% and 75% of the cardiac cycle. Based on the ECG signal, the scans were reconstructed at two different phases of the cardiac cycle: the one that corresponds to the best phase, that is, the phase with the least motion, and the one with the strongest motion artifacts. Due to the limited z-coverage of the CT scanner, the heart cannot be imaged in one shot. Therefore, each scan is composed of typically three to five stacks that are acquired sequentially at different z-positions. Here, each stack was reconstructed on a $512 \times 512 \times 58$ voxel grid with a voxel spacing of about $0.35 \text{ mm} \times 0.35 \text{ mm} \times 0.6 \text{ mm}$. To compensate for coronary artery motion, each stack was processed as described in Section 2.A.1. Due to the absence of a ground truth, the results were compared against the conventional PAMoCo that was implemented according to reference.¹⁹

It has to be noted that the Deep PAMoCo network is currently designed to process only 15 z-slices at a time. Therefore, the simulations as well as the measurements need to be processed sequentially, that is, the PARs need to be divided into several $128 \times 128 \times 15$ subvolumes with a certain overlap of 0–14 slices. In this study we chose the overlap to be 14 slices, that is, the final motion-compensated reconstruction is composed of the central slices of the corresponding subvolumes.

3. RESULTS

3.A. Simulation study

Motion-compensated reconstructions were performed for 5000 simulated samples. The corresponding data were generated as described in Section 2.B. Reconstructions of four exemplary samples are shown in Fig. 5. Here, the Deep PAMoCo is able to remove motion artifacts efficiently and

yields CT reconstructions that are almost equal to the ground truth.

A qualitative evaluation was performed in terms of the MAE_{CA} according to Eq. (4). In particular, the MAE_{CA} was evaluated with respect to the contrast c , the radius r and the displacement d of the coronary artery. Here, the latter is defined as:

$$d = \int |\partial_t s(t)| dt, \quad (5)$$

where $s(t)$ represents the simulated motion vector for the projection at time point t . The corresponding results are shown in Fig. 6 (contrast), Fig. 7 (radius), and Fig. 8 (displacement). While high errors of up to 300 HU can be observed if no correction is applied, the application of the Deep PAMoCo reduces these errors to about 25 HU. Furthermore, it can be seen that the Deep PAMoCo performs equally well for different contrasts, radii, and displacements.

3.B. Clinical cases

The Deep PAMoCo was applied to 13 clinical cardiac CT scans. Due to the absence of a ground truth the Deep PAMoCo results were evaluated qualitatively by a comparison to the uncorrected reconstruction as well as to the conventional PAMoCo of Hahn et al. Axial slices as well as sliding thin-slab multiplanar reformations (STS MPR) along the coronary artery are shown in Figs. 9–11 for three exemplary patient cases. Considering the best phase, which is at 70% of the cardiac cycle for the three cases, the initial reconstruction without correction already shows reasonable image quality. Nevertheless, motion artifacts are clearly visible in some axial slices. Here, both the PAMoCo and the Deep PAMoCo are able to reduce these artifacts almost completely. However, it has to be noted that both motion compensation approaches introduce a slight blurring which can be attributed to the resampling operation of the spatial transformer as

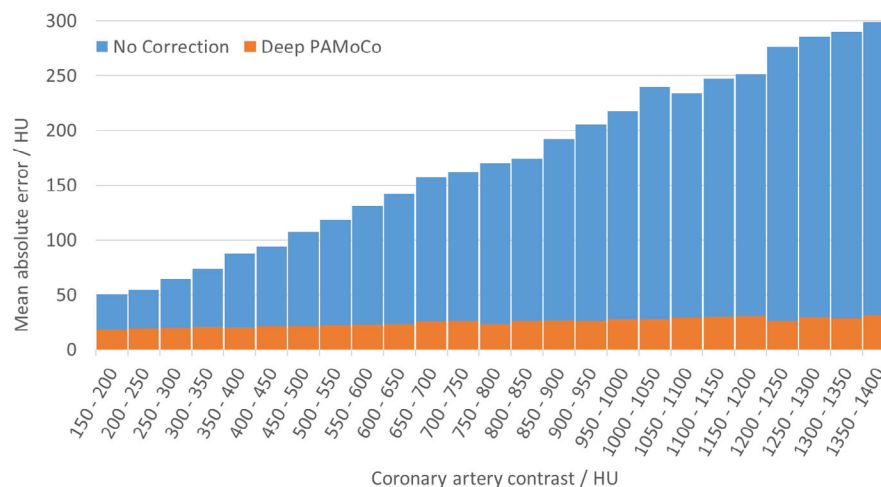


FIG. 6. Mean absolute error along the coronary artery with respect to an ideal motion compensation for different contrasts. Errors are averaged for contrast intervals of 50 HU. [Color figure can be viewed at wileyonlinelibrary.com]

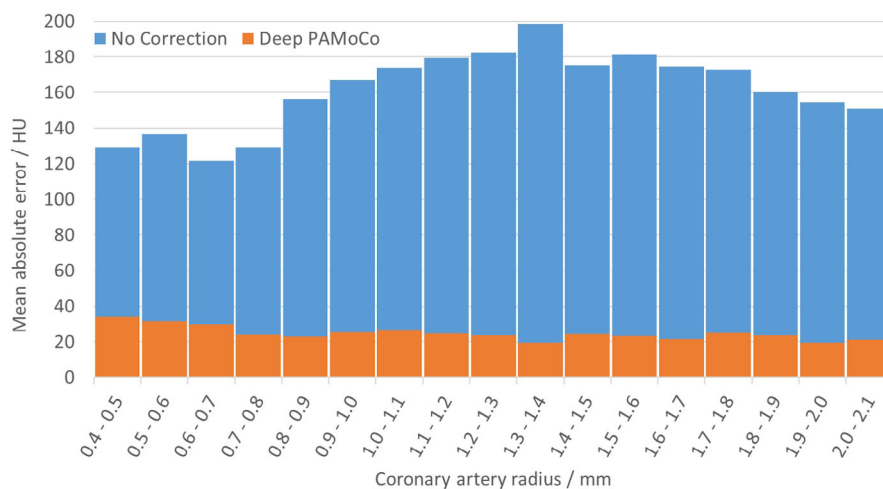


FIG. 7. Mean absolute error along the coronary artery with respect to an ideal motion compensation for different radii. Errors are averaged for radius intervals of 0.1 mm. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/mp.14827)]

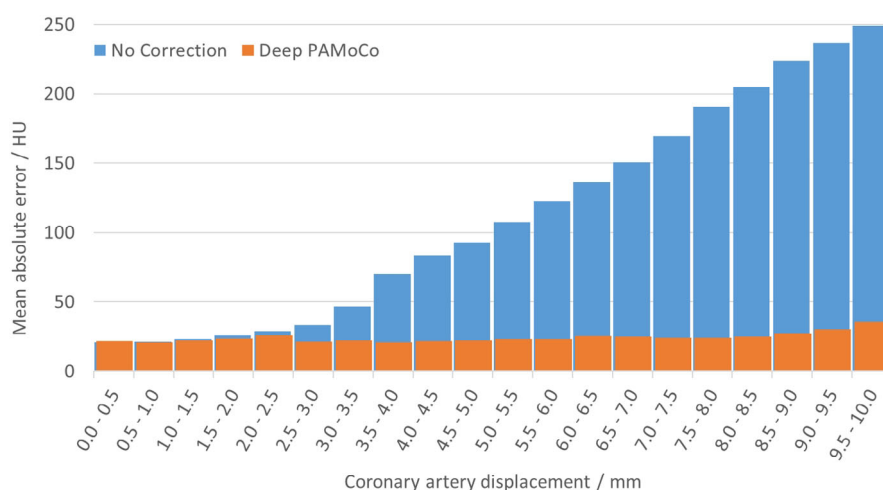


FIG. 8. Mean absolute error along the coronary artery with respect to an ideal motion compensation for different displacements. Errors are averaged for displacement intervals of 0.5 mm. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/mp.14827)]

well as the additional forward- and backprojection described in Section 2.A.1.

A more severe impact of motion can be observed for the reconstructions at 50% of the cardiac cycle.

Although this phase is typically not considered for diagnosis, it can be seen as a proxy for a scan with a bad timing of the acquisition window (best phase not available) or for a scan of a patient with a high heart rate.

In this case, strong artifacts are visible in the axial slices as well as in the STS MPR (right column of Figs. 9–11) if no correction is applied. In particular, the coronary arteries appear narrowed, faded, or disrupted in several areas which make these reconstructions unsuitable for a proper diagnosis of potential coronary artery diseases. Similar to the reconstructions at 70% of the cardiac cycle, motion compensation clearly improves image quality. However, in some slices the conventional PAMoCo fails to completely compensate for the motion artifacts. Here, the Deep PAMoCo shows a superior

performance and provides CT reconstructions with almost no remaining artifacts.

4. DISCUSSION AND CONCLUSION

In this manuscript, the Deep PAMoCo is proposed to compensate for coronary artery motion. Its basic principle is inspired by the work of Hahn et al. that is based on the PAR concept. Here, MVFs are estimated and used to deform a set of PARs such that they represent the same motion state. However, in contrast to their approach, the Deep PAMoCo does not apply a conventional optimization strategy to do so, but uses a deep neural network instead. This leads to a boost of computational performance by about two orders of magnitude and allows to process a cardiac CT scan in about 15 s.

Compared to existing deep learning-based approaches for coronary artery motion compensation, the main novelty of

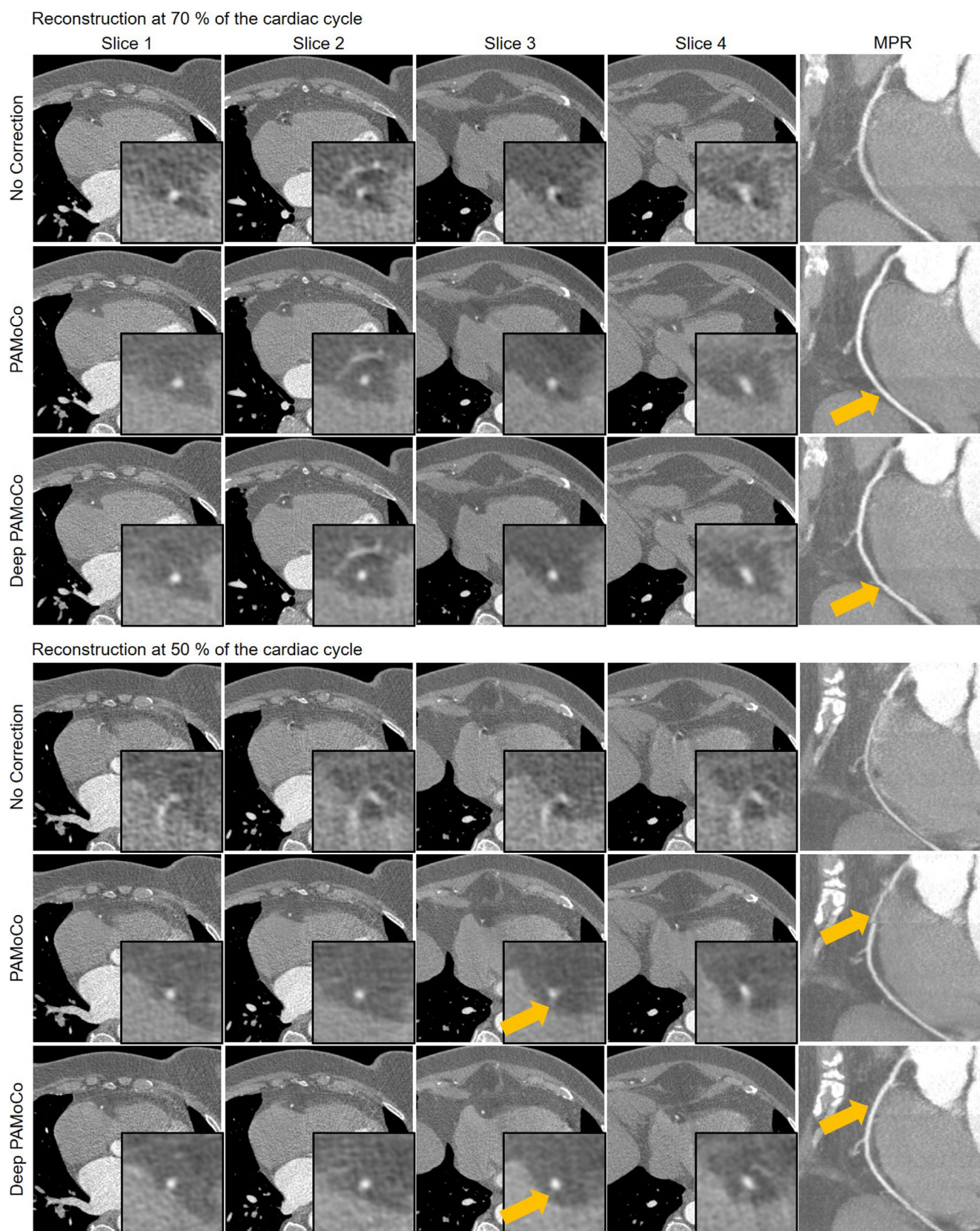


FIG. 9. Patient case 1: Heart rate = 69 bpm, tube voltage = 120 kV, cardiac CT reconstructions ($C = 0$ HU, $W = 1000$ HU) at 70% of the cardiac cycle (top) and 50% of the cardiac cycle (bottom). The first four columns show four different axial slices without correction (first rows), conventional PAMoCo (middle rows), and Deep PAMoCo (bottom row). The last column shows a 6-mm sliding thin-slab maximum intensity projection of a multiplanar reformation (STS MPR) of the coronary artery. The arrows indicate major differences between the PAMoCo and the Deep PAMoCo. It has to be noted that the CT reconstruction is composed of several stacks. Therefore, discontinuities may arise in the MPR at the stack boundaries. [Color figure can be viewed at wileyonlinelibrary.com]

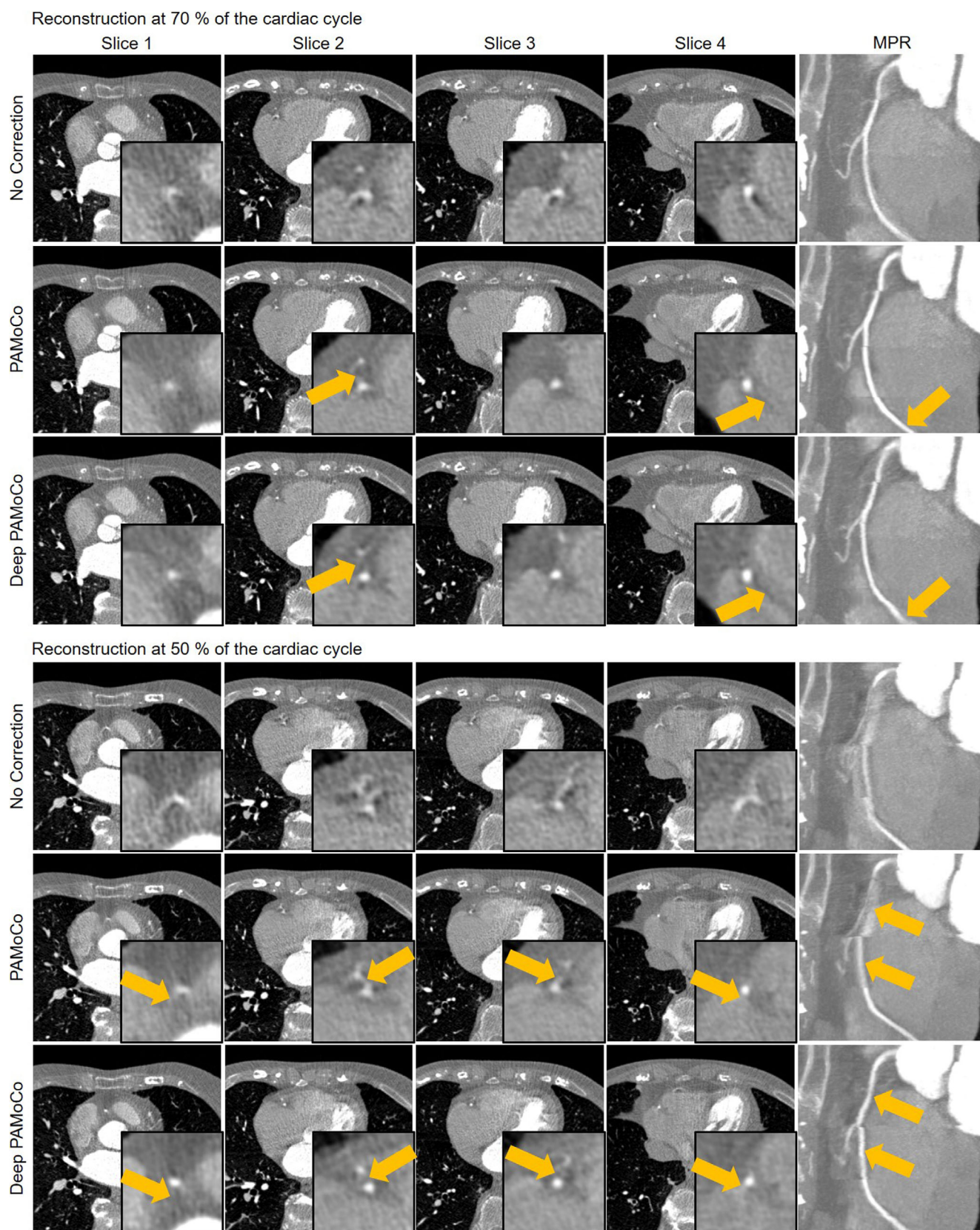


FIG. 10. Patient case 2: Heart rate = 66 bpm, tube voltage = 80 kV, cardiac CT reconstructions ($C = 0 \text{ HU}$, $W = 1800 \text{ HU}$) at 70% of the cardiac cycle (top) and 50% of the cardiac cycle (bottom). The first four columns show four different axial slices without correction (first rows), conventional PAMoCo (middle rows), and Deep PAMoCo (bottom row). The last column shows a 6-mm sliding thin-slab maximum intensity projection of a multiplanar reformation (STS MPR) of the coronary artery. The arrows indicate major differences between the PAMoCo and the Deep PAMoCo. It has to be noted that the CT reconstruction is composed of several stacks. Therefore, discontinuities may arise in the MPR at the stack boundaries. [Color figure can be viewed at wileyonlinelibrary.com]

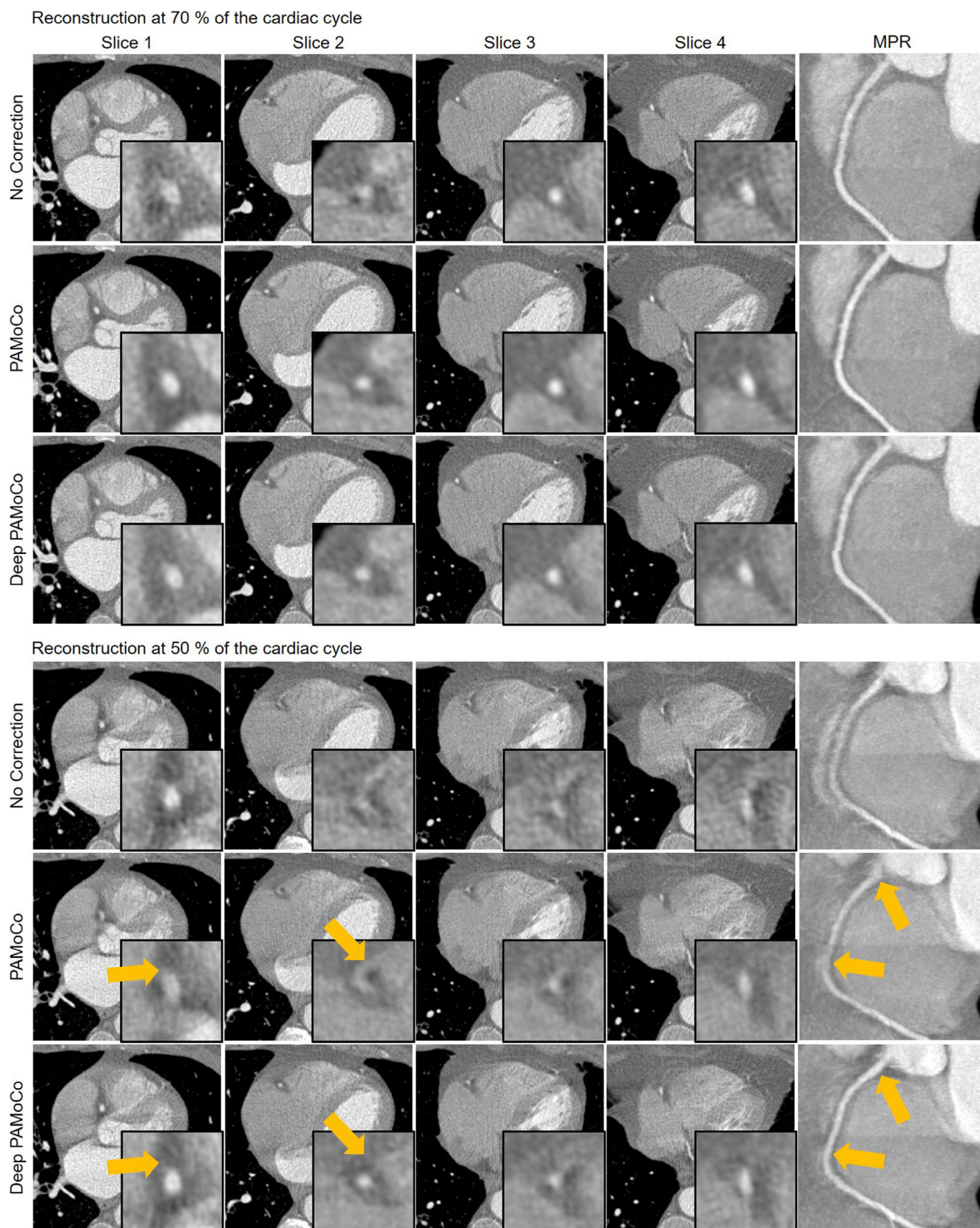


FIG. 11. Patient case 3: Heart rate = 70 bpm, tube voltage = 100 kV, cardiac CT reconstructions ($C=0\text{ HU}$, $W=1500\text{ HU}$) at 70% of the cardiac cycle (top) and 50% of the cardiac cycle (bottom). The first four columns show four different axial slices without correction (first rows), conventional PAMoCo (middle rows), and Deep PAMoCo (bottom row). The last column shows a 6-mm sliding thin-slab maximum intensity projection of a multiplanar reformation (STS MPR) of the coronary artery. The arrows indicate major differences between the PAMoCo and the Deep PAMoCo. It has to be noted that the CT reconstruction is composed of several stacks. Therefore, discontinuities may arise in the MPR at the stack boundaries. [Color figure can be viewed at wileyonlinelibrary.com]

the Deep PAMoCo is the use of the PAR concept, the non-linear motion model, and the spatial transformer module. Here, the former two contributions put a strong constraint on the possible deformations to be applied. This helps to make the Deep PAMoCo more traceable and to avoid the introduction of artificial structures. The spatial transformer module, which internally applies the motion model to the PARs, enables an efficient end-to-end training. This is particularly useful as the loss function can be evaluated in image domain. In that way the loss directly accounts for motion artifacts and introduces an implicit weighting of the parameters of the motion model with respect to their improvement of image quality. This makes it potentially easier to train more complex motion models and may be particularly beneficial for further studies that aim to apply the motion compensation to the entire heart or the cardiac valves. Furthermore, it allows to employ unsupervised training strategies which may for instance be used to map data of single-source CT (low temporal resolution) to reconstructions of a dual-source CT (high temporal resolution) via a generative adversarial network.²⁸

In this study, however, the Deep PAMoCo was trained in a supervised setup using simulated training data. The performance of the trained network was evaluated for simulations as well as patient measurements. Here, the simulation study demonstrated that the Deep PAMoCo generalizes well to different start angles of the gantry, different diameters of the coronary artery, different contrasts as well as different displacements of the coronary artery. In any case, the Deep PAMoCo provided motion-compensated CT reconstructions that were almost equal to the corresponding ground truth. Similar results were obtained for the measurements. This indicates that the simulation, and especially the motion model with three control points that was used to generate the training data, is sufficiently complex to model real coronary artery motion even for large motion amplitudes. In particular, this is demonstrated by comparing the results for the best phase to the phase with the strongest motion artifacts. Here, no major differences can be observed when applying the Deep PAMoCo.

This suggests that diagnostic image quality can even be achieved for cases where the standard reconstruction shows severe motion artifacts, for example, due to an inappropriate timing of the acquisition window or due to high heart rates and contraindications to beta-blockade. Beyond that, the Deep PAMoCo could even open new options to perform cardiac CT scans completely without heart rate control drugs or ECG-gating.

In contrast, there is a stronger dependency on the motion phase for the conventional PAMoCo. Here, more severe motion artifacts are compensated less efficiently. A possible explanation for this behavior is that the stronger the motion artifacts are, the more likely it is for the optimization to get trapped in a local minimum of the MAM.

Finally it has to be noted that the current implementation of the Deep PAMoCo has still some limitations, most of which are related to the low z-coverage. As the Deep PAMoCo does not allow to take more than 15 z-slices as

input, cardiac CT scans must be divided into subvolumes, which are then processed sequentially. Potentially, this strategy may cause inconsistencies between adjacent subvolumes. Furthermore, a low z-coverage is not favorable in terms of processing time. Since the MVFs also act in z-direction, the subvolumes need to have a certain overlap to avoid boundary effects. Here, we choose to use the maximum overlap of 14 slices which requires to call the Deep PAMoCo $N - 15$ times to process a CT reconstruction with N slices. Obviously, the straight forward solution to this problem is to train the Deep PAMoCo with more z-slices. However, as the coronary artery is not aligned to the z-axis, this would also require to increase the axial dimension and we would face memory issues with our current hardware.

Despite these limitations, which will be subject to further research, it can be concluded that the Deep PAMoCo provides an efficient approach that outperforms state-of-the-art coronary artery motion compensation approaches in terms of processing time as well as accuracy. Therefore, it may help to further improve the diagnosis of coronary artery diseases even if the corresponding CT scans are corrupted by strong motion artifacts.

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CONFLICT OF INTEREST

The authors have no conflicts to disclose.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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